

TAREA

1) $4 + 2 \times 5$

$$4 + 10 = \underline{14}$$

2) $23 \times 2 / 5 =$

$$\begin{array}{r} 23 \\ \times 2 \\ \hline 46 \end{array}$$

$$\begin{array}{r} 46 \overline{) 5} \\ 10 \\ \hline 19 \\ 10 \\ \hline 9 \\ 10 \\ \hline 0 \end{array} = \underline{9,2}$$

$$3) 6 + 2 \times 3 - 4/2 =$$

$$\begin{array}{r} 6 + 6 - 4/2 \\ 6 + 6 - 2 = \underline{10} \end{array}$$

$$4) 5 \times 5 (5 + (6 - 2) + 1)$$

$$5 \times 5 (5 + 4 + 1)$$

$$5 \times 5 (10)$$

$$25 \times 10 = \underline{250}$$

$$5) 7 - 6/3 + 2 \times 3/2 - 4/2 \times 2$$

$$7 - 2 + 6/2 - 2 \times 2$$

$$7 - 2 + 3 - 4$$

$$5 - 1$$

$$\underline{4}$$

$$\begin{array}{r} -6/3 \\ 0-2 \end{array}$$

$$\begin{array}{r} -4/2 \\ 0-2 \end{array}$$

$$\begin{array}{r} 6/2 \\ 0-3 \end{array}$$

$$6) 5 \div 2 = \underline{1}$$

$$\begin{array}{r} 5/2 \\ 1-2 \end{array}$$

$$7) 45/5 \div 2$$

$$9 \div 2$$

$$\underline{1}$$

$$\begin{array}{r} 45/5 \\ 0-9 \end{array}$$

$$\begin{array}{r} 9/2 \\ 1-4 \end{array}$$

$$8) 3 + 5 \times (10 - (2 + 4))$$

$$3 + 5 \times (10 - 6)$$

$$3 + 5 (4)$$

$$3 + 20$$

$$\underline{23}$$

$$9) 3 + 5 \times (10 - 6)$$

$$3 + 5 (-16)$$

$$3 - 80$$

$$\underline{-77}$$

$$10) 3.5 + 5.09 - 14.0/40$$

$$3.5 + 5.09 - 0.35$$

$$8.59 - 0.35$$

$$\underline{8.24}$$

$$\begin{array}{r} 1400/40 \\ 200-0.35 \\ 0 \end{array}$$

$$+ 3.5$$

$$5.09$$

$$\underline{8.59}$$

$$8.59$$

$$- 0.35$$

$$\underline{8.24}$$

$$11) \cancel{3.5} + 5.09 - \cancel{3.5} + 8.59 - \cancel{3.5}$$

$$\begin{array}{r} 5.09 - 3.5 + 8.59 \\ 1.59 + 8.59 \\ \hline 10.18 \end{array}$$

$$\begin{array}{r} 4 \\ 5.09 \\ - 3.50 \\ \hline 1.59 \end{array} \quad \begin{array}{r} 1.1 \\ 1.59 \\ + 8.59 \\ \hline 10.18 \end{array}$$

$$12) 2.1 * (1.5 + 30 * 4.1)$$

$$2.1 * (1.5 + 12.30)$$

$$2.1 * (13.80)$$

$$\underline{28.98}$$

$$\begin{array}{r} * 3.0 \\ 4.1 \\ \hline 3.0 \end{array}$$

$$\begin{array}{r} 12.30 \\ 120.0 \\ \hline 123.0 \end{array}$$

$$\begin{array}{r} 12.30 \\ + 1.5 \\ \hline 13.80 \end{array}$$

$$\begin{array}{r} 13.80 \\ * 2.1 \\ \hline 13.80 \\ 276.0 \\ \hline 289.80 \end{array}$$

$$m) 2.1 * (1.5 - 12.3)$$

$$2.1 * (-10.8)$$

$$\underline{-22.68}$$

$$\begin{array}{r} 1 \\ -12.3 \\ - 1.5 \\ \hline 10.8 \end{array}$$

$$\begin{array}{r} 1 \\ -10.8 \\ * 2.1 \\ \hline -10.8 \\ 216 \\ \hline -22.68 \end{array}$$

$$n) (5 - 3\% (4 - 2)) + 3 * 2$$

$$(5 - 3\% 2) + 3 * 2$$

$$(5 - 1) + 3 * 2$$

$$4 + 6$$

$$\underline{10}$$

$$\begin{array}{r} 3 \overline{) 2} \\ 10 \overline{) 15} \\ 0 \end{array}$$

$$o) 7 \times 4 - 2 + 5 - (2 + 2 \times 5) \times 2 - 3$$

$$7 \times 4 - 2 + 5 - (2 + 10) \times 2 - 3$$

$$7 \times 4 - 2 + 5 - (12) \times 2 - 3$$

$$28 - 2 + 5 - (12) \times 2 - 3$$

$$28 - 2 + 5 - 24 - 3$$

$$26 + 5 - 24 - 3$$

$$31 - 24 - 3$$

$$7 - 3$$

$$\underline{4}$$

$$\begin{array}{r} 31 \\ -24 \\ \hline 07 \end{array}$$

$$p) 8 + 7 \times 3 + 4 \times 6$$

$$8 + 21 + 4 \times 6$$

$$29 + 4 \times 6$$

$$29 + 24$$

$$\underline{53}$$

$$q) -2^3 = \underline{-8}$$

$$r) (33 + 3 \times 4) / 5$$

$$(33 + 12) / 5$$

$$45 / 5$$

$$\underline{9}$$

$$1) = 9$$

$$s) 2^2 \times 3$$

$$4 \times 3$$

$$\underline{12}$$

$$t) 3 + 2 \times (18 - 4^2)$$

$$3 + 2 \times (18 - 16)$$

$$3 + 2 \times (2)$$

$$3 + 4$$

$$\underline{7}$$

$$u) 16 \times 16 - 3 \times 2$$

$$256 - 6$$

$$\underline{250}$$

$$\begin{array}{r} 16 \\ \times 16 \\ \hline 96 \\ 16 \\ \hline 256 \end{array}$$

$$16$$

$$96$$

$$16$$

$$256$$

$$v) 5/2 + 20 \bmod 6$$

$$2.5 + 20 \bmod 6$$

$$2.5 + 2$$

$$\underline{4.5}$$

$$5/2$$

$$102.5$$

$$0$$

$$20/6$$

$$23$$

$$w) 4 \times 6/2 - 15/2$$

$$24/2 - 15/2$$

$$12 - 7.5$$

$$\underline{4.5}$$

$$24/2$$

$$012$$

$$-15/2$$

$$10-7.5$$

$$0$$

$$12$$

$$-7.5$$

$$12.5$$

$$7.5$$

$$\underline{4.5}$$

$$x) 5 \times 15/2 / (4-2)$$

$$5 \times 15/2/2$$

$$75/2/2$$

$$37,5/2$$

$$18,75$$

$$\begin{array}{r} 2 \\ 15 \\ \hline 75 \end{array}$$

$$\begin{array}{r} 75 \overline{) 2} \\ 15 \quad 37,5 \\ \hline 10 \\ 0 \end{array}$$

$$\begin{array}{r} 375 \overline{) 2} \\ 17 \quad 18,75 \\ \hline 15 \\ 10 \\ 0 \end{array}$$

$$y) 10 + 5 \times (2^3)$$

$$10 + 5 \times (8)$$

$$10 + 40$$

$$50$$

$$2) 100 / 4 / 5 \times 2$$

$$25 / 5 \times 2$$

$$5 \times 2$$

$$10$$

$$\begin{array}{r} 25 \overline{) 5} \\ 0 \quad 5 \end{array}$$

$$aa) 15 \% 4 + 2 \times 3$$

$$3 + 6$$

$$9$$

$$\begin{array}{r} 15 \overline{) 4} \\ 3 \quad 3 \end{array}$$

$$BB) 25 + (10 - 2^2) \times 4/2$$

$$25 + (10 - 4) \times 4/2$$

$$25 + (6) \times 4/2$$

$$25 + 24/2$$

$$25 + 12$$

$$37$$

$$\begin{array}{r} 25 \\ + 12 \\ \hline 37 \end{array}$$

$$\begin{array}{r} 24 \overline{) 2} \\ 04 \quad 12 \\ \hline 0 \end{array}$$

$$cc) (8+2) \times (5-3)^2 / 10$$

$$10 \times (5-3)^2 / 10$$

$$10 \times 2^2 / 10$$

$$10 \times 4 / 10$$

$$40 / 10$$

$$4$$

$$\begin{array}{r} 10 \\ \times 4 \\ \hline 40 \end{array}$$

$$\begin{array}{r} 40 \overline{) 10} \\ 0 \quad 4 \end{array}$$

$$\begin{aligned} y &= V+V=V \\ V+f &= F \\ f+f &= F \end{aligned}$$

$$\begin{aligned} 0 &= V+V=V \\ V+f &= V \\ f+f &= F \end{aligned}$$

Punto 2 Taller matemáticas

$$a=10 \quad b=20 \quad c=30 \quad x=15 \quad y=10 \quad z=5$$

a) $a+b > c$

$$10+20 > 30 = \underline{F}$$

g) $b+(2-a)/5 < a \times c \times 2$

$$20+(2-10)/5 < 10 \times 30 \times 2$$

$$20+8/5 < 10 \times 30 \times 2$$

$$20+(1.6) < 600$$

$$18.4 < 600 = \underline{V}$$

b) ~~$2 \times 2 \times 5$~~ $a-b < c$

$$10-20 < 30 = \underline{V}$$

c) $a-b=c$

$$10-20=30 = \underline{F}$$

d) $a \times b \neq c$

$$10 \times 20 \neq 30 = \underline{V}$$

h) $8=16$ $a \neq 4$ y $4 < 1$

$$\underline{8=16} \quad (V) \quad (F)$$

$$(F) \quad 0 \quad y$$

$$(F) \quad (F) = \underline{(F)}$$

e) $a \times c - b \times a < b \times 5$

$$10 \times 30 - 20 \times 10 < 20 \times 5$$

$$300 - 200 < 100$$

$$100 < 100 = \underline{F}$$

i) $(4 \times 3 < 6, 0, 3 > 5-2) y 3+2 < 12$

$$(F) \quad 0 \quad (F)$$

$$(F) \quad y \quad (V)$$

$$= \underline{(F)}$$

f) $c \times a - b < a + b + c$

$$30 \times 10 - 20 < 10 + 20 + 30$$

$$300 - 20 < 10 + 20 + 30$$

$$280 < 60 = \underline{F}$$

j) $(x > y) y (z < x)$

$$\underline{15 > 10} y \underline{5 < 15}$$

$$(V) y (V)$$

$$= \underline{(V)}$$

k) $(x+y=25) \circ (z > x)$

$$(15+10=25) \circ (5 > 15)$$

$$(V) \quad 0 \quad (F)$$

$$= \underline{(V)}$$

l) $!(x \neq y+z)$

$$!(15 \neq 10+5)$$

$$!(15 \neq 15)$$

$$= \underline{(V)}$$

$$m) (X \% 2 == 0) \vee (Y \% 2 == 0)$$

$$(15 \% 2 == 0) \vee (10 \% 2 == 0)$$

$$1 == 0$$

$$(F) \quad 0 \quad 0 \% == 0$$

$$(V)$$

$$= (V)$$

$$\begin{array}{r} 15 \overline{) 2} \\ 1 \end{array}$$

$$\begin{array}{r} 10 \overline{) 2} \\ 0 \end{array}$$

$$n) (X - 2 >= Y) \vee (Z * 2 == Y)$$

$$(15 - 5 >= 10) \vee (5 * 2 == 10)$$

$$\underline{10 >= 10} \vee \underline{10 == 10}$$

$$(V)$$

$$(V)$$

$$(V)$$

3) Deducir el valor de las siguientes expresiones: $A=5$; $B=25$; $C=10$

$$X \leftarrow A + B + C$$

$$X = 5 + 25 + 10$$

$$X = 30 + 10$$

$$X = \underline{40}$$

$$X \leftarrow A + B * C$$

$$X = 5 + 25 * 10$$

$$X = 5 + 250$$

$$X = \underline{255}$$

$$X \leftarrow A + B / C$$

$$X \leftarrow 5 + 25 / 10$$

$$X \leftarrow 5 + 2,5$$

$$X \leftarrow \underline{7,5}$$

$$X \leftarrow A + B \backslash C$$

$$X \leftarrow 5 + 25 \backslash 10$$

$$X \leftarrow 5 \backslash 2,5$$

$$X \leftarrow \underline{7,5}$$

$$X \leftarrow A + B \bmod C$$

$$X = 5 + 25 \% 10$$

$$X = 5 + 5$$

$$X = \underline{10}$$

$$\begin{array}{r} 25 \overline{) 10} \\ 5 \end{array}$$

$$5 \overline{) 2}$$

$$X \leftarrow (A + B) \backslash C$$

$$X = (5 + 25) \backslash 10$$

$$X = 30 \backslash 10$$

$$X = \underline{3}$$

$$\cancel{X \leftarrow A + B / C}$$

$$X$$

$$X$$

$$X \leftarrow A + (B / C)$$

$$X = 5 + (25 / 10)$$

$$X = 5 + 2,5$$

$$X = \underline{7,5}$$

$$\begin{array}{r} 25 \overline{) 10} \\ 0 \end{array}$$

$$0 \overline{) 25}$$

$$0$$

4) $A = 5$ $A = 86$
 $B = A + 4$ $B = 111$
 $A = A + 1$
 $B = A - 5$

5) $A = 3$ $A = 8842$ $A = 56$
 $B = 5$ $B = 51614$ $B = 14$
 $A = B + 3$
 $B = A * 2$
 $B = B - 2$
 $A = B * 3$
 $A = A + B$

6) $P \leftarrow -10$ $P = 163620$ $B) 1. x \leftarrow 2$ $x = 224$ $24 | 8$
 $7) Q \leftarrow -20$ $Q = 2010$ $2) y \leftarrow -3$ $y = 30$ 03
 $3) P \leftarrow P + Q$ $4) x \leftarrow x \wedge y$ $z = 8$
 $4) Q \leftarrow P - Q$ $5) y \leftarrow x \% z$

7) $a = 8, b = 3, c = -5$

a. $a + b + c$ $b = 2 * b + 3 * (a - c)$ $d = a / b$
 $8 + 3 + -5$ $2 * 3 + 3 * (8 - (-5))$ $8 / 3$
 $11 - 5$ $2 * 3 + 3 * 13$ $= 22$
 6 $6 + 39$ 2
145

d) a mod b e) a / c f) a mod c
 $8 \% 3$ $8 / -5$ $8 / 5$
 $= 2$ $= -16$ $301,6$
 22 0 31

$$\textcircled{g) \quad a * (b/c) \quad 30/-5} \\ 8 * (3/-5) \quad 00,6 \\ 8 * 0,6 \\ = \underline{48}$$

$$h) \quad a * (b/c) \\ 8 * (3/-5) \\ 8 * 0,6 \\ = \underline{48}$$

$$i) \quad (a * c) \bmod b \\ 40 \% 3 \\ = \underline{-1}$$

$$\begin{array}{r} 8 \\ \times 5 \\ \hline 40 \\ 40 / 3 \\ 10 \quad 13 \\ 1 \end{array}$$

$$j) \quad a * (c \bmod b) \\ 8 * (2) \quad -5 / 3 \\ 8 * (2) \quad 2 \quad 1 \\ = \underline{16}$$

8) Calcular el valor de la Variable

$$a) \quad \text{Valor} \leftarrow 4.0 * 5 = \underline{20.0}$$

$$B) \quad X \leftarrow 3.0 \\ Y \leftarrow 2.0 \quad = \underline{7.0} \\ \text{Valor} \leftarrow X \wedge Y - Y \\ \text{Valor} \leftarrow 3.0 \wedge 2.0 - 2.0$$

$$c) \quad \text{Valor} \leftarrow 5 \quad V = \underline{15} \\ X \leftarrow 3 \quad X = 3 \\ \text{Valor} \leftarrow \text{Valor} * X$$

$$g) \quad a \leftarrow 4 \quad A = \underline{4} \\ b \leftarrow a \quad b = \underline{4} \\ b \leftarrow a + 3 \quad b = \underline{7}$$

$$b = a \leftarrow 5 \quad A = \underline{6} \\ b \leftarrow a + b \quad b = \underline{11} \\ a \leftarrow a + 1 \\ b \leftarrow a - 5$$

$$c) \quad a \leftarrow 3 \quad A = \underline{10} \\ b \leftarrow 20 \quad b = \underline{23} \\ c \leftarrow a + b \quad c = \underline{23} \\ b \leftarrow a + b \\ a \leftarrow b - c$$

d) $a \leftarrow 10$ $A = \underline{10}$ $\underline{5}$

$b \leftarrow 5$ $b = \underline{5}$

$a \leftarrow b$

$b \leftarrow a$